

DEMONSTRATION OF PROPOSED FEATURES

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

The Rising Waters project demonstrates how Machine Learning can be used to predict flood risk based on rainfall and environmental parameters. The application provides an interactive Streamlit interface where users can enter data and instantly receive flood risk predictions.

Proposed Features

1. Interactive User Interface

The application provides a clean and easy-to-use Streamlit dashboard for entering rainfall and weather parameters.

2. Rainfall Data Input

Users can enter rainfall values to analyze flood risk.

3. Weather Parameter Analysis

The system accepts temperature and humidity values to improve prediction accuracy.

4. Machine Learning Prediction

The trained XGBoost model analyzes the input values and predicts whether the flood risk is High or Low.

5. Real-Time Prediction

Prediction results are generated instantly after clicking the **Predict Flood Risk** button.

6. Accurate Results

The trained model provides reliable predictions based on historical rainfall and weather data.

7. User-Friendly Dashboard

The dashboard is designed with simple navigation and an attractive interface for better user experience.

8. Easy Deployment

The application is deployed using Streamlit Cloud, allowing users to access it from a web browser.

Demonstration Steps

1. Open the Rising Waters application.
 2. Enter rainfall, temperature, and humidity values.
 3. Click **Predict Flood Risk**.
 4. The Machine Learning model processes the input.
 5. The application displays the flood risk prediction.
 6. Users can test different values to observe prediction changes.
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Expected Output

- Fast prediction response
 - High prediction accuracy
 - Easy-to-understand results
 - Smooth user interaction
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Benefits

- Early flood prediction
 - Supports disaster management
 - Easy to use
 - Fast processing
 - Reliable results
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Conclusion

The Rising Waters application successfully demonstrates all proposed features. The system provides accurate flood prediction using Machine Learning and offers a simple, efficient, and user-friendly solution for flood risk analysis.